Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 1 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 1 : SLO â€“ 1 | Stacks â€” Introduction

I. Fill in the Blanks

• A stack is a LINEAR (LIFO) data structure.
• The two main operations of a stack are PUSH and POP.
• The operation that adds an element to a stack is called PUSH.
• The operation that removes the top element from the stack is called POP.
• The operation that returns the top element without removing it is called PEEK.
• If a stack is full and a push operation is attempted, it is called OVERFLOW.
• If a stack is empty and a pop operation is attempted, it is called UNDERFLOW.
• Stacks are usually implemented using ARRAYS or LINKED LISTS.
• In array implementation of stack, the top element is accessed using the TOP index.
• In function calls, the system uses a stack to store RETURN ADDRESSES (activation records).
• The initial value of top in an empty stack array implementation is usually -1.
• Stack is used in POSTFIX (Reverse Polish) notation to evaluate expressions without parentheses.
• A stack is also used in DEPTH-FIRST traversal of trees.
• The time complexity of push and pop operations in a stack is O(1).
• Reversing a string using stack works by PUSHING each character and then POPPING them.